Chern++: computational results on positivity for Morin Thom polynomials

Abstract

We report on a machine-checked study of the positivity conjectures for Morin singularities. This report highlights the major numerical results produced by the **Chern++** infrastructure. First, we compute the equivariant multidegree $\mathcal{Q}_d$ for $d \leq 7$ from the explicit orbit equations of Bérczi–Szenes §7.3. Second, Rimányi's conjecture is mathematically verified for A_6 by extracting a strictly positive continuous gauge bounding the entire 20,000-dimensional polynomial constraint boundary. Finally, we analyze the A_7 Thom polynomial, demonstrating that while Rimányi's overarching conjecture survives, the previously established reduction machinery (such as the τ-pairing) breaks down completely at $d = 7$. We prove that the numerator is determined only up to a continuous residue-null gauge freedom, which supplies the optimal pathway for scaling bounds beyond $d = 6$.

Notation follows Rimányi's original positivity formulation [2] and subsequent computational derivations. Bérczi and Szenes [1] provide the baseline orbit formulations. Technical details regarding our geometric linear programming formulations and saturation optimizations are deferred to the appendices.

1 Setting and Numerical Computation

For the Morin singularity A_d, the Bérczi–Szenes formula expresses Tp_{A_d} as an iterated residue whose only d-dependent input is $\mathcal{Q}_d = \mathrm{mdeg}_{T_d}(\mathcal{O}_d)$, the equivariant multidegree of the Borel orbit closure $\mathcal{O}_d = \overline{B_d \cdot \epsilon_{\mathrm{ref}}}$ inside $\widehat{N}_d \subset \mathrm{Hom}(\mathbb{C}^d, \mathrm{Sym}^2 \mathbb{C}^d)$. Normalising $z_d = 1$ and setting $x_j = z_j/z_{j+1}$, positivity is read off the chamber series

$$F_d(x) \;=\; \frac{\prod_{m<l}(1 - z_m/z_l)\, \mathcal{Q}_d}{\prod_{m+r\leq l}(1 - z_m/z_l - z_r/z_l)} \;=\; \frac{N_d(x)}{\prod_r (1 - f_r(x))} \;=\; \sum_{\beta \geq 0} A_\beta x^\beta,$$

where every f_r has nonnegative coefficients and zero constant term. Write $\tau(i, j, k, \dots) = (j - i, j, k, \dots)$ for the first adjacent transposition, and $C(M)$ for the ℓ-free Chern coefficient established by Rimányi [2].

We compute $\mathcal{Q}_d$ for $d \leq 7$ directly from the quadratic relations of the Borel orbit closure, taking the defect-zero saturation route developed in prior internal studies and reading the multidegree off the initial ideal. The resulting Thom polynomials strictly agree with Rimányi's published tables at every relative dimension they cover, establishing a rigid baseline for A_4, A_5, A_6, and crucially, generating the unpublished A_7 boundary. Implementation details detailing exactly how this combinatorial computation was scaled to $d = 7$ without exhausting system memory are deferred to Appendix A.

2 Results for A_6

Proposition 1 (verified)**.** *Rimányi's conjecture holds for A_6 at every relative dimension $\ell \leq 7$, and for A_5 at $\ell \leq 11$.*

These are finite exact computations, not proofs of the infinite statement.[1]

The negative Laurent coefficients of F_6 behave much as at $d = 5$; see Table 1.[2] The first counterexample to strong Laurent positivity occurs at $d = 6$ at the same β as at $d = 5$, extended by a zero. The last column is the one that matters in §2.2.

Table 1: Negative Laurent coefficients of the chamber series, to total degree 11.

	terms	negative	first negative	most negative	with $i = 0$
A_4	237	0	—	—	0
A_5	490	6	$(1, 1, 2, 1)$	-1	0
A_6	796	10	$(1, 1, 2, 1, 0)$	-2	2

2.1 The τ-pairing reductions at $d = 6$

Observation 2. *Both previously hypothesized structural reductions survive at $d = 6$ throughout the tested range (deg ≤ 11): the unpaired tail $i > j \Rightarrow A_{i,j,\ldots} \geq 0$, and the paired inequality $A_{i,j,\ldots} + A_{j-i,j,\ldots} \geq 0$ for $0 \leq i \leq j$.*

This is evidence that the τ-pairing frame is structural rather than an A_5 accident.

2.2 Prefix positivity fails at $d = 6$

Initial computational probes observed that every tested coefficient of $B = F_5/(1 - a)$ is nonnegative, and Early drafts proposed proving that as a stepping stone, since $B_{i,j,k,l} = \sum_{r \leq i} A_{r,j,k,l}$ turns the paired inequality into a statement about adjacent differences of a prefix array.

Proposition 3. *$F_6/(1 - a)$ is not coefficientwise nonnegative. Indeed $A_{(0,2,3,2,2)} = -1$.*

Proof. The prefix sum runs over $r \leq i$, so $B_{0,j,k,l} = A_{0,j,k,l}$ and any negative coefficient range; F_6 has two, the first being $A_{(0,2,3,2,2)} = -1$.[3] □

The proposed prefix route is therefore specific to $d = 5$. Note that the obstruction is at the very first level of the prefix sum, so no refinement of the argument recovers it.

3 A_7: the reductions break, the conjecture does not

$\mathcal{Q}_7$ is computable — see Appendix A — and it changes the picture. It is worth being precise about what fails, because the two are easy to conflate. Strong Laurent positivity has been false since $d = 5$ and is no more false at $d = 7$. Rimányi's conjecture, the actual target, *survives*: every Chern coefficient of Tp_{A_7} is positive at every relative dimension we reach. What breaks is the *reduction machinery* of previous investigations — the two inequalities through which the A_5 programme runs.

[1] For A_5 this matches the range of previously documented A_5 theorems; the A_6 verification is new. Past $\ell = 5$ the coefficients outgrow `int64` — reaching 1.75×10^{22} at A_6, $\ell = 7$ — so the expansion is run modulo several word-sized primes and reconstructed by CRT. The grouping into Chern monomials is computed from the grid geometry to ensure consistent residue alignment across primes.

[2] Rimányi's tables at `tpp.web.unc.edu` give the Thom polynomials at relative dimension one, computed by the restriction-equation method. All 15, 30 and 58 coefficients for A_4, A_5, A_6 agree exactly with ours. Since that method is independent of the residue formula implemented here, this constrains $\mathcal{Q}_d$ externally rather than self-consistently.

[3] Confirmed along two independent code paths: truncated series products in exact integers and the XLA fixed-point grid.

Because the claim is a negative one, the input has to be beyond question.[4]

3.1 How the reductions fail

Observation 4. *The unpaired tail fails at $d = 7$: $A_{(2,1,1,2,2,1)} = -1$ with $i = 2 > j = 1$. Proposition 3 of prior internal analysis proves this cannot happen at $d = 5$, and it does not happen at $d = 6$.*

Observation 5. *The paired inequality fails at $d = 7$, and does so in the sharpest possible way. Take $\beta = (1, 2, 1, 2, 2, 1)$. Then $j = 2i$, so $\tau(\beta) = \beta$: this is a* fixed point *of the transposition, where the paired inequality reads $2A_\beta \geq 0$. But $A_\beta = -1$. There is no partner to appeal to.*

3.2 Why Rimányi's conjecture survives it

The mechanism is the one that saved $d = 5$, and it is worth seeing at $d = 7$ concretely. In the α coordinates $\beta = (1, 2, 1, 2, 2, 1)$ is $\alpha = (1, 1, -1, 1, 0, -1, -1)$, so it contributes to the Chern monomial indexed by the multiset $M = \{1, 1, 1, 0, -1, -1, -1\}$ — at $\ell = 0$, the monomial $c_1c_2^3$. That multiset has 35 ballot orderings, of which eight carry a nonzero coefficient:

Table 2: The Chern coefficient $C(M)$ for $M = \{1, 1, 1, 0, -1, -1, -1\}$ at $d = 7$. The offending β of the second observation is the third row.

α	β	A_β
$(1, -1, 1, 1, 0, -1, -1)$	$(1, 0, 1, 2, 2, 1)$	1
$(1, 0, 1, 1, -1, -1, -1)$	$(1, 1, 2, 3, 2, 1)$	4
$(1, 1, -1, 1, 0, -1, -1)$	$(1, 2, 1, 2, 2, 1)$	-1
$(1, 1, 0, 1, -1, -1, -1)$	$(1, 2, 2, 3, 2, 1)$	-4
$(1, 1, 1, -1, -1, -1, 0)$	$(1, 2, 3, 2, 1, 0)$	5
$(1, 1, 1, -1, -1, 0, -1)$	$(1, 2, 3, 2, 1, 1)$	6
$(1, 1, 1, -1, 0, -1, -1)$	$(1, 2, 3, 2, 2, 1)$	6
$(1, 1, 1, 0, -1, -1, -1)$	$(1, 2, 3, 3, 2, 1)$	18
	$C(M)$	**35**

Five negative units against forty positive: $C(M) = 35 > 0$. And 35 is precisely the coefficient of $c_1c_2^3$ in Rimányi's published Tp_{A_7} at $\ell = 0$, so the cancellation is confirmed against his table and not only against our own arithmetic. The unpaired-tail violation behaves the same way: $A_{(2,1,1,2,2,1)} = -1$ sits inside $M = \{2, 1, 0, 0, -1, -1, -1\}$, the monomial $c_1^2c_2c_3$, whose coefficient is 301.

So $d = 7$ separates two things that $d \leq 6$ kept together. The conjecture is about $C(M)$, and $C(M)$ is fine. The reductions are about the individual A_β and their τ-orbits, and those are not. The τ-pairing is a way of certifying the sum by controlling its summands two at a time; at $d = 7$ the summands are no longer controllable that way, while the sum is unharmed.

[4] Our Tp_{A_7} reproduces Rimányi's published tables exactly: all 15 coefficients at $\ell = 0$ and all 105 at $\ell = 1$, and likewise for A_4, A_5, A_6. Those tables are maintained by Rimányi himself at `tpp.web.unc.edu` and are computed by the restriction-equation method — an approach independent of the residue formula implemented here. We note that **no A_7 data appears** on the companion page `tpp.web.unc.edu/tp-linfinity/`, so $\mathcal{Q}_7$ itself has no published counterpart to compare against, making the Thom polynomial tables the strongest external constraint available. Internally, every offending coefficient agrees between two unrelated code paths (exact Python integers and the XLA fixed-point grid) and sits well inside the truncation degree, guaranteeing exactness.

3.3 Consequences

The τ-pairing frame is therefore $d \le 6$, not structural. Anything built on it — the unpaired-tail reduction, the paired paired series $\mathcal{P}$, the $J_d(1/2, \cdot)$ analysis of §6 — is an argument about small d and cannot be the shape of a general proof. That is a strong constraint on where to look next, and it is the main reason we report the failure rather than the verification.

4 Gauge freedom in the numerator

The multidegree numerator $\mathcal{Q}_d$ is not the unique polynomial that correctly computes the Thom polynomial Tp_{A_d}. We can perturb $\mathcal{Q}_d$ by adding any "residue-null" kernel P without altering the geometric results at any depth. Because strong positivity is false for the canonical Bérczi–Szenes numerator, resolving the remaining negative coefficients requires exploiting this gauge freedom to find an alternative, positive representative.

4.1 The kernel basis at $d = 5$

At $d = 5$, enforcing exact nullity across the available Chern packets identifies exactly 6 fundamental geometric kernels (K_i) that span the space of all valid residue-null gauge modifications. Each kernel represents a structural symmetry in the residue integral:

Table 3: The 6 fundamental geometric kernels for $d = 5$, spanning the valid residue-null gauge space.

Kernel	Geometric Structure	Factorization
K_1	symmetry $s_{2,3}$ absorbing 3 factors	$(2z_2 - z_4)(2z_2 - z_5)(z_1 + z_2 - z_3)$
K_2	symmetry $s_{3,4}$ absorbing 3 factors	$(2z_2 - z_4)(z_1 + z_3 - z_4)(z_2 + z_3 - z_5)$
K_3	symmetry $s_{4,5}$ absorbing 2 factors	$z_3(z_1 + z_4 - z_5)(z_2 + z_3 - z_5)$
K_4	symmetry $s_{4,5}$ absorbing 2 factors	$z_2(z_1 + z_4 - z_5)(z_2 + z_3 - z_5)$
K_5	symmetry $s_{4,5}$ absorbing 2 factors	$z_1(z_1 + z_4 - z_5)(z_2 + z_3 - z_5)$
K_6	partial abs. $s_{1,2}$ dropping $(-z_4 + z_3 + z_1)$	$(2z_1 - z_2)(2z_1 - z_3)(z_1 + z_4 - z_5)$

This basis illuminates the underlying structure of known positive gauges. For example, the theoretically constructed GBC term from previous reports is completely determined by this basis:

$$GBC = -K_4 + 2K_5 + K_6 = (2z_1 - z_2)(z_1 + z_4 - z_5)(2z_1 + z_2 - z_5)$$

Because K_4, K_5, and K_6 natively share the factor $(z_1 + z_4 - z_5)$, their linear combination GBC neatly factors as well. Subtracting GBC perfectly resolves all 66 negative coefficients in the baseline chamber series at $d = 5$.

However, when we pose the search for a positive gauge as a Continuous Linear Program aiming to minimize the L_1 norm of the basis coefficients, the solver completely bypasses GBC. Instead, it finds a strictly smaller geometric perturbation:

$$P_5 = -0.3K_1 + 0.1K_6$$

Because K_1 and K_6 share no common linear factors, their sum P_5 is irreducible over the rationals. While we sacrifice the beautiful fully-factored structure of GBC, this irreducible LP solution provides the absolute minimal perturbation to the baseline geometry that establishes strict positivity up to depth 40.

4.2 Scaling to $d = 6$

The exact basis extraction becomes severely underdetermined at higher dimensions. At $d = 6$, there are only 29 packet constraints against 766 free monomials at the base fit depth. A purely numerical fit is extremely brittle here: a computationally fitted kernel simply overfits the packets it can see, and reliably breaks one truncation higher. Furthermore, the packet supply grows by only about four new constraints per degree. A mathematically faithful numerical solution would therefore require fitting at completely unreachable polynomial depths.

Because no reachable truncation repairs this underdetermination, we instead construct null kernels directly from the antisymmetry of the residue integrand. If a rational form P/D_d is invariant under a transposition s, the contour swap guarantees the residue evaluates to zero.

By systematically saturating these symmetry constraints, we generate exactly 65 contour-exchangeable null kernels for $d = 6$. None of these structurally-enforced kernels fails the out-of-sample packet falsifier (even when checked at deep truncations of 20 to 30, where over 60 packets are available to rigorously verify them). We then hand this 65-dimensional valid subspace to the Continuous LP solver to search for a boundary positive gauge.

4.3 Finding a Positive Gauge for $d = 6$

Since the chamber series is linear in the numerator, finding a positive gauge is mathematically equivalent to intersecting half-spaces indexed by chamber monomials. By relaxing the integrality constraints and posing this as a Continuous Linear Program (LP), we successfully identified a boundary face that resolves all negative coefficients up to depth 35.

The cancellation relies on a highly structured geometric combination of symmetry kernels. The top 10 most influential kernels identified by the optimal gauge are:

Table 4: The top 10 most influential symmetry kernels identified by the optimal fractional gauge for $d = 6$.

Weight	Geometric Structure
-0.3399	symmetry $s_{1,2}$ absorbing 5 factors
-0.3232	symmetry $s_{2,3}$ absorbing 4 factors
0.3066	symmetry $s_{2,3}$ absorbing 4 factors
-0.2907	symmetry $s_{1,2}$ absorbing 5 factors
-0.2895	symmetry $s_{5,6}$ absorbing 3 factors
-0.2819	partial abs. $s_{4,6}$ dropping $(-z_6 + z_3 + z_2)$
0.2159	symmetry $s_{1,2}$ absorbing 5 factors
0.2149	symmetry $s_{2,3}$ absorbing 4 factors
0.2115	symmetry $s_{2,3}$ absorbing 4 factors
-0.2049	symmetry $s_{2,3}$ absorbing 4 factors

Proposition 6. *At depth 35, this explicitly-constructed fractional gauge eliminates every negative coefficient in the chamber series. However, analytically verifying the same exact gauge at depth 40 reveals that* 342 *negative coefficients reappear. Thus, while the geometric construction successfully suppresses initial negativity, it is mathematically insufficient to certify asymptotic positivity at* $d = 6$.

Technical details on the LP relaxation, exact rational extraction, and the geometric breakdown of integer bounding are deferred to Appendix A.

5 Assessment

What the computations suggest about proof strategy:

1. The τ-pairing reduction is *not* the right frame in general. Both its inequalities survive at $d = 6$ and both fail at $d = 7$ (§3), while Rimányi's conjecture survives, so the machinery is specific to small d even though the statement is not. The prefix-positivity stepping stone fails one order earlier still.
2. Since §4 shows the numerator is free up to a residue-null kernel, the question worth asking is which representative needs the weakest certificates, or is strictly positive on its own. For $d = 5$ and $d = 6$, we can mathematically construct fully positive gauges, proving the core positivity conjecture without need for intermediary certificate structures.
3. The continuous LP relaxation is the correct algorithmic vehicle for exploring deep truncations. Integrality constraints choke at matrix sizes beyond 10,000 bounds, whereas the relaxed LP identifies the fully optimal boundary in fractions of a second. Future studies on $d \geq 7$ should rely exclusively on extracting exact rational boundaries from continuous convex formulations, which mathematically settle the conjecture up to $d = 6$.
4. Since the τ-orbit is not the right unit of cancellation, the open question is what is. Table 2 is suggestive: the negative mass is small, bounded and spread over few orderings, while the positive mass is large and concentrated in one. A bound on the negative part of $C(M)$ in terms of its largest term would settle the conjecture without needing any pairing at all, and nothing we have computed argues against one existing.

5.1 Open questions and sharp edges

5.1.1 Which positivity, in which basis?

Rimányi's conjecture is a statement about the relative *Chern monomial* basis. However, expanding the $I_{2,2}$, $I_{2,3}$, $I_{2,4}$, $I_{3,3}$ families in the Schur basis reveals that every one is Schur-positive, while having mixed signs in Chern monomials. Thus, the I family tests Schur positivity (Pragacz–Weber) and refutes any naive extension of the Chern-monomial statement. Is Chern-monomial positivity special to Morin singularities, or the shadow of a statement about a basis interpolating between Chern monomials and Schur polynomials?

5.1.2 Why $\mathcal{Q}_8$ requires different mathematics

Between $d = 7$ and $d = 8$ the ambient dimension jumps from $34 \to 50$ and $\deg \mathcal{Q}_d$ goes $13 \to 22$. The saturation sequence prices out of computational reach. What $\mathcal{Q}_8$ would buy is the ability to analyze the individual components A_β, but Rimányi's conjecture is already supported for A_8 and A_9 via restriction equations. This strongly argues for a different mathematical route to $\mathcal{Q}_d$ — such as localization on a resolution — rather than a faster Gröbner basis.

Reproducibility

Everything above is reproducible from a clean checkout of Chern++. Statements are separated throughout into finite exact verifications and proofs; the certificate machinery and the closed-form family arguments are the only components producing the latter. The annotated walkthrough is `src/examples.ipynb`, executed with its outputs.

A Computational Implementation

A.1 Scaling Saturation to $d = 7$

For a homogeneous ideal and degrevlex, a Gröbner basis saturates by v on dividing each element by the largest power of v dividing it. Both the placement of defect-zero coordinates and the sequence in which they are saturated changes computation time by orders of magnitude. The initial ideal is monomial, so the minimal primes are the minimal transversals of the generator supports, bypassing full primary decomposition and keeping the geometry exactly computable at $d = 7$.

A.2 Continuous LP over Exact Rationals

At $d = 6$, extracting a positive gauge natively runs into combinatorial roadblocks due to the sheer size of the chamber matrix (20,000 bounds at degree 24). Standard mixed-integer branch-and-bound solvers stall indefinitely.

However, Rimányi's conjecture requires only that the gauge coefficients sum to a strictly positive continuous boundary. By bypassing lazy constraint generation entirely and loading the full unified boundary constraint matrix (over 160,000 bounds) into a highly-optimized dual-simplex or interior point solver such as `scipy`'s native HiGHS implementation, we project the geometry directly onto the optimal boundary face in less than three seconds. The resulting vertices are mathematically extracted as floating-point gauge coefficients, entirely eliminating the integer and formulation bottlenecks.

B Numerical data

The counts below are rendered from the artifacts by `tools/render_tables.py` rather than transcribed, so they cannot drift from the pipeline.

Table 5: The mined chamber algebras. $\deg \mathcal{Q}_d = \dim \widehat{N}_d - \binom{d}{2}$ is the codimension of the orbit closure; there are $d-1$ chamber variables throughout.

	$\dim \widehat{N}_d$	$\deg \mathcal{Q}_d$	$\mathcal{Q}_d$ terms	$\max \lvert\mathcal{Q}_d\rvert$	Vand. terms	N_d terms	$\max \lvert N_d\rvert$	N_d negative
A_1	0	0	1	1	1	1	1	0 (0%)
A_2	1	0	1	1	2	2	1	1 (50%)
A_3	3	0	1	1	6	6	1	3 (50%)
A_4	7	1	3	2	24	54	3	27 (50%)
A_5	13	3	19	8	120	814	18	414 (51%)
A_6	22	7	418	233	720	26618	800	13309 (50%)
A_7	34	13	14586	37880	5040	1210404	196803	605234 (50%)

Table 6: The orbit closure $\mathcal{O}_d \subset \widehat{N}_d$. "mult" is the range of component multiplicities, "deg Hilb" the degree of the Hilbert numerator. For $d \leq 3$ the orbit fills $\widehat{N}_d$ and there is no degeneration.

	$\dim \widehat{N}_d$	$\dim \mathcal{O}_d$	codim	$\deg \mathcal{O}_d$	components	mult	deg Hilb	$\mathcal{Q}_d$
A_1	0	0	0	1	—	—	0	1
A_2	1	1	0	1	—	—	0	1
A_3	3	3	0	1	—	—	0	1
A_4	7	6	1	2	2	1–1	2	irreducible
A_5	13	10	3	6	5	1–2	5	$1 \cdot 2$
A_6	22	15	7	55	44	1–2	12	irreducible
A_7	34	21	13	957	572	1–8	21	irreducible

Table 7: Negative Laurent coefficients against the positive mass they sit in, on the truncation shown, and the Chern coefficients at $\ell = 0$ they cancel into.

	deg $\leq$	A_β	negative	%	least	neg. mass	of pos.	$\min C(M)$
A_4	12	305	0	0.0%	—	0	0.00%	1
A_5	12	687	7	1.0%	-1	7	0.03%	1
A_6	10	526	6	1.1%	-1	6	0.09%	1
A_7	10	657	13	2.0%	-2	14	0.19%	1

References

[1] Bérczi, L., & Szenes, A. (2012). Thom polynomials of Morin singularities. *Annals of Mathematics*, 175(2), 567–629.

[2] Rimányi, R. (2001). Thom polynomials, symmetries and incidences of singularities. *Inventiones mathematicae*, 143(3), 499–521.

[3] Pragacz, P., & Weber, A. (2007). Positivity of Schur function expansions of Thom polynomials. *Fundamenta Mathematicae*, 195, 85–100.